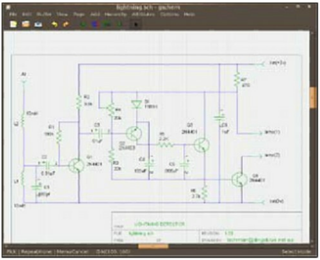

Figure 2: A schematic drawn using gschem

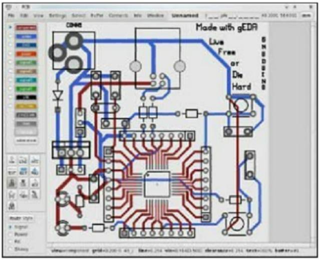

Figure 3: The AT90USB162 mini development board layout made with *pcb*

to draw schematics for an electronic design—gschem is for that purpose.

You can invoke it from the command line by typing *gschem*. It will open a graphical application with all the menus and buttons necessary to draw a schematic. You'll find gschem very similar to other EDA programs. Besides, you can easily master it using the tutorials available at the gEDA documentation site (*http://geda.seul.org/wiki*). This tool will give you all the options for wires, component pins, components, etc, that you require for your PCB design.

Since all your files are saved in the ASCII format, scripting languages can easily manipulate these. Thanks to this feature, there are several symbol generation tools that have been made using Python and Perl, besides the footprint generators made using Python. This is in stark contrast to proprietary EDA programs, which lock in users by not supporting ASCII file formats.

Connectivity information using gnetlist

The schematic you prepared using gschem needs a netlist. The term 'netlist' means a net connections list, where the 'net' is the connection between components. The netlist file contains the list of connections between the component pins. The tool for this task is *gnetlist*. In simple words, *gnetlist* is a graphics-to-text converter tool. Netlist information is mainly used by circuit simulators to simulate the schematic you have drawn—as the input to layout programs that typically hold information about each component's PCB footprint, as well as connectivity information between all component pins.

A wide range of around 20 different file formats can be outputted by the netlister. Among these file formats, the SPICE format is popular. These file formats can be used to simulate electrical circuits that are popular among electrical and electronics engineering students worldwide.

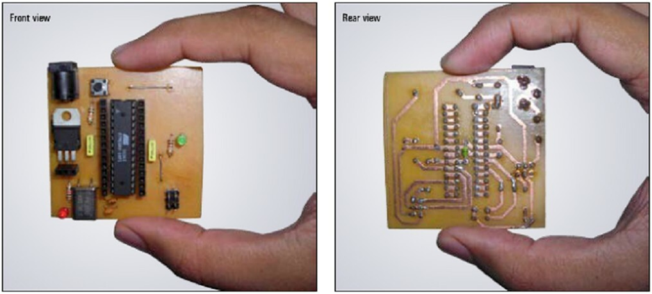

Figure 4: A home-made PCB by the author (ATMega8 mini development board)